

POWER BI PROJECT REPORT

E-Commerce Sales & Profitability Analytics

Power BI Business Analytics Project

Tools: Excel | Power Query | Power BI | DAX

1. Project Overview

This project develops an interactive Power BI analytics solution for an e-commerce business to evaluate **sales performance, profitability, customer behaviour, product performance, discounts, returns, refunds and cancellations**.

The objective was to transform raw transactional data into a reliable analytical model and a multi-page management dashboard that supports **performance monitoring, problem identification and business decision-making**.

Dataset

The analysis uses six tables:

- **Orders** - transactional sales and order information
- **Customers** - customer demographics and segments
- **Products** - product, category, sub-category and pricing information
- **Returns** - return and refund transactions
- **Regions** - city-to-region mapping
- **Date** - calendar and time-analysis dimension

The final model contains **2,500 orders, 350 customers, 80 products and 165 return records**.

2. Business Objectives / Questions

The dashboard was designed to answer the following business questions:

Sales Performance

- What are total sales, profit, margin and order volume?
- How does sales performance vary over time?
- Which categories, regions and channels contribute most to sales?
- Which payment modes are most used?
- What is the average order value?

Customer Analytics

- How many active customers are there?
- What is the new vs. repeat customer pattern?
- Which customer segments generate the most sales?

- Who are the highest-value customers?
- Which states generate the most customer profit?

Product & Profitability

- Which sub-categories and products generate the most sales?
- Which products generate the most profit?
- How do product tiers compare?
- How do discounts relate to profitability?

Returns & Operations

- What is the return rate and refund value?
- Which categories have the highest returns?
- What are the major return reasons?
- What is the cancellation rate?
- Does cancellation vary by channel?

These questions follow the business-question framework in the project guide.

3. Data Cleaning & Power Query

Power Query was used to prepare the source data before modelling.

Key steps included:

- Reviewed table structure, row counts, columns and data types.
 - Converted dates and numerical fields to appropriate data types.
 - Checked missing values, errors and duplicates.
 - Validated financial calculations including **Gross Sales, Discount Amount, Sales Amount and Profit**.
 - Checked CustomerID and ProductID referential integrity.
 - Validated the Date table and month sorting.
 - Validated the cleaned data against business rules and source totals.
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4. Data Model & Relationships

A **star-schema approach** was implemented with Orders as the primary sales fact table and Returns as the second fact table.

Main relationships

- Customers → Orders
- Customers → Returns
- Products → Orders
- Products → Returns

- Date → Orders
- Date → Returns
- Regions → Customers

Relationships use **one-to-many, single-direction filtering** wherever applicable.

The Date table was configured as the report's date dimension, with Year, Quarter, Month and YearMonth attributes. The Returns table contains both **OrderDate and ReturnDate**. The active reporting relationship uses OrderDate, while ReturnDate is retained for return-date analysis.

5. DAX Measures / KPI Definitions

Sales & Profitability

- **Total Sales** - total value of sales transactions
- **Total Profit** - total profit generated
- **Profit Margin** - profit as a percentage of sales
- **Average Order Value** - average sales value per order
- **Total Orders** - distinct number of orders
- **Total Quantity** - total units sold
- **Total Discount** - total discount amount
- **Average Discount %** - average discount percentage

Customers

- Total Customers
- Active Customers
- Repeat Customers
- New Customers
- Average Revenue per Customer

Returns & Operations

- Total Returns
- Return Rate
- Total Refunds
- Net Sales
- Net Profit
- Cancellation Rate

6. Dashboard Overview

6.1 Executive Overview

Provides a management-level summary of:

- Total Sales
- Total Profit
- Total Orders
- Active Customers
- Profit Margin
- Net Sales
- Return Rate
- Monthly sales and profit
- Orders by channel
- Profit by category
- Returns by reason
- Sales by geography
- Discount vs profit relationship

Interactive filters include **Year, Quarter, Region and Category**.

6.2 Sales Analysis

Focuses specifically on revenue generation and sales mix.

Includes:

- Total Sales
- Average Order Value
- Average Discount %
- Sales by Category
- Sales by Quarter
- Sales by Region
- Sales by Channel
- Sales by Payment Mode

Year and Quarter filters allow focused sales analysis.

6.3 Customer Analytics

Focuses on customer value and behaviour.

Includes:

- Active Customers

- Repeat Customers
 - New Customers
 - Orders over time
 - Top 10 customers by sales
 - Average Order Value by segment
 - Sales by customer segment
 - Profit by state
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6.4 Product & Profitability

Evaluates product-level and category-level profitability.

Includes:

- Total Profit
 - Profit Margin
 - Sales by sub-category
 - Profit by category
 - Sales and profit by product tier
 - Top products by profit
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6.5 Returns & Operations

Evaluates post-sale performance and operational issues.

Includes:

- Return Rate
 - Total Refunds
 - Cancellation Rate
 - Returns by category
 - Refunds by category
 - Returns by reason
 - Orders by status
 - Cancelled orders by channel
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6.6 Management Insights

A decision-oriented page combining financial and operational indicators.

Includes:

- Total Sales
 - Total Profit
 - Return Rate
 - Profit Margin
 - Category Performance & Risk matrix
 - Product-level sales/profit comparison
 - Product-level order analysis
 - Discount vs Profit Margin by Product Tier
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7. Business Insights

1. Strong overall profitability

The business generated approximately **₹22.46M in sales and ₹6.75M in profit**, resulting in an overall **30.0% profit margin**.

Interpretation: The business is generating a substantial profit contribution relative to sales.

Action: Management should protect profitable categories and monitor factors such as discounting, product mix and returns that can affect margins.

2. Fashion is the leading sales category

Fashion generated approximately **₹5.2M**, followed by Beauty at approximately **₹4.8M** and Sports at approximately **₹4.7M**.

Interpretation: Fashion is the largest contributor to category sales.

Action: Management can monitor Fashion's product mix and profitability while identifying opportunities to strengthen lower-performing categories.

3. Sales are concentrated across the major regions

West and North contribute approximately **30.7% and 30.4%** of sales respectively, while South contributes approximately **28.4%**.

Interpretation: These three regions account for the majority of revenue, while East contributes a substantially smaller share.

Action: Management can investigate the drivers of the regional difference and evaluate whether distribution, customer acquisition or product availability explains the gap.

4. Mobile App is the strongest sales channel

The Mobile App leads the channels in sales/orders, followed by Website, while Marketplace contributes the smallest share.

Interpretation: The business has stronger activity through its direct digital channels.

Action: Management should monitor mobile-app performance and evaluate the reasons for the lower Marketplace contribution before changing channel investment.

5. Consumer customers drive the majority of sales

The Consumer segment contributes approximately **61.3% of total sales**, compared with **23.7% Corporate** and **15.0% Small Business**.

Interpretation: Revenue is concentrated primarily in the Consumer segment.

Action: Customer strategies should protect the existing Consumer base while evaluating opportunities to increase Corporate and Small Business value.

6. The customer base is predominantly repeat-active

The dashboard shows **349 active customers**, including **346 repeat customers** and **3 customers classified as new under the project's measure definition**.

Interpretation: The current customer base is strongly represented by repeat purchasing behaviour.

Action: Management should maintain retention-focused initiatives while separately monitoring customer acquisition.

7. Footwear is the strongest sub-category

Footwear generates approximately **₹2.7M in sales**, ahead of Makeup and Kitchenware.

Interpretation: Footwear is the strongest sales-generating sub-category in the dashboard.

Action: Management should monitor Footwear's inventory, pricing and product-level profitability to sustain its contribution.

8. Returns are relatively limited but have identifiable causes

The overall return rate is **2.21%**, with refund value of approximately **₹574.5K**. The four major return reasons are relatively distributed: Wrong Product (**27.27%**), Customer Changed Mind (**26.06%**), Damaged (**24.24%**) and Late Delivery (**22.42%**).

Interpretation: No single return reason dominates the return mix.

Action: Improvements should address product accuracy, packaging/product quality and delivery processes rather than focusing on only one cause.

9. Cancellations are a significant operational area

The order-status analysis shows approximately **19.4% cancelled orders**, with Delivered orders representing approximately **80.6%**.

Interpretation: Cancellation is a more significant operational issue than the return rate.

Action: Management should investigate cancellation drivers by channel, particularly the channels with higher cancellation volumes.

10. Product tiers show similar profitability levels

Profit margins across Budget, Standard and Premium tiers are relatively close, around **30%**, while discount levels are also broadly similar.

Interpretation: The dashboard does not show a large profitability gap between the three product tiers.

Action: Management should evaluate individual products and discount levels rather than assuming that one product tier is inherently more profitable.

8. Business Recommendations

1. Protect high-performing categories

Maintain availability and monitor profitability of high-sales categories such as Fashion, Beauty and Sports.

2. Review cancellation drivers

Investigate cancellation patterns by channel and identify operational causes such as fulfilment, payment or customer-order issues.

3. Reduce avoidable returns

Focus on product accuracy, packaging quality and delivery performance to reduce Wrong Product, Damaged and Late Delivery returns.

4. Strengthen customer retention

The high proportion of repeat customers indicates the importance of maintaining customer experience and repeat-purchase behaviour.

5. Diversify regional performance

Investigate the lower contribution from East relative to the other major regions and identify opportunities for growth.

6. Use discounts selectively

Evaluate discounts at product/category level and monitor their effect on profit rather than increasing promotions solely to increase sales.

7. Focus on profitable products

Prioritize products that combine strong sales with strong profit contribution, while reviewing products with weaker profitability.

9. Conclusion

The Power BI solution converts raw e-commerce transaction data into a structured analytical reporting system covering **sales, profitability, customers, products, discounts, returns and operations**.

The six dashboards provide different levels of analysis, from executive performance monitoring to detailed customer, product and operational analysis.

The analysis indicates that the business generates **₹22.46M in sales with ₹6.75M profit and a 30.0% margin**, while key management opportunities include **maintaining profitable categories, monitoring cancellations, controlling returns, strengthening customer retention and applying discounts selectively**.

The project demonstrates an end-to-end analytics workflow covering **data cleaning, modelling, DAX, dashboard development, validation and business interpretation**.
